

## Year 7 · Chapter 9 · The number plane: answers (teacher copy)

### 9.01 Grid references on maps (booklet p. 3)

#### WE DO examples (teacher)

- Ex 1a: the lake
- Ex 1b: the library
- Ex 2a: D9
- Ex 2b: B2, C2, B3, C3
- Ex 3a: e.g. walk to Gauss Rd, west to Euclid St, then north
- Ex 3b: it is a convention so everyone reads it the same way

#### Easy (drill)

- |          |           |        |
|----------|-----------|--------|
| 1. A5    | 11. Park  | 21. B2 |
| 2. D1    | 12. Bus   | 22. C4 |
| 3. E4    | 13. Post  | 23. H2 |
| 4. G5    | 14. Oval  | 24. H6 |
| 5. F6    | 15. Wharf | 25. A1 |
| 6. H6    | 16. Farm  | 26. E2 |
| 7. Zoo   | 17. Pool  | 27. B6 |
| 8. Gym   | 18. Fire  | 28. F6 |
| 9. Bank  | 19. G3    | 29. H5 |
| 10. Cafe | 20. E5    | 30. F1 |

#### Medium (drill)

- B5
- E6
- H3
- E5 (Club)
- F5
- G4 (Oval)
- F3
- E5 (Club)
- C5
- G3 (Cafe)

#### Challenging

- own map
- own directions

#### Extension

- C5
- 18 squares

#### Exit ticket

- Level 1: stadium
- Level 2: F2
- Level 3: student centre (H7)

### 9.02 Points in the first quadrant (booklet p. 9)

#### WE DO examples (teacher)

- Ex 1a: plotted
- Ex 1b: plotted on the axes
- Ex 2a: P(4, 1), Q(2, 5)
- Ex 2b: No: (2, 5) is 2 across, 5 up
- Ex 3a: rectangle
- Ex 3b: (0, 0)

#### Easy (drill)

- |       |            |            |
|-------|------------|------------|
| 1. 6  | 8. 7       | 15. (4, 6) |
| 2. 10 | 9. 8       | 16. (4, 5) |
| 3. 4  | 10. 4      | 17. (2, 6) |
| 4. 1  | 11. 10     | 18. (6, 6) |
| 5. 12 | 12. 0      | 19. (1, 8) |
| 6. 12 | 13. (7, 8) | 20. (6, 0) |
| 7. 6  | 14. (0, 0) | 21. (8, 8) |

22. (2, 8)

23. (1, 2)

24. (0, 3)

#### Medium (drill)

- |      |      |       |
|------|------|-------|
| 1. B | 5. C | 9. K  |
| 2. J | 6. E | 10. L |
| 3. M | 7. D | 11. A |
| 4. F | 8. H | 12. G |

#### Challenging

- (1, 6)
- x-axis points have  $y = 0$ ; y-axis points have  $x = 0$

#### Extension

- (3, 5)
- 24 square units

#### Exit ticket

- Level 1: 7
- Level 2: on the y-axis, 5 up
- Level 3: (1, 5)

### 9.03 The number plane (booklet p. 15)

#### WE DO examples (teacher)

- Ex 1a: 1st top right, 2nd top left, 3rd bottom left, 4th bottom right
- Ex 1b: 3rd
- Ex 2a: A(2, 3), B(-3, 1), C(-2, -4), D(4, -2)
- Ex 2b: 4th
- Ex 3a: a zigzag line
- Ex 3b: e.g. (-2, 3); e.g. (0, 5)

#### Easy (drill)

- |        |              |              |
|--------|--------------|--------------|
| 1. 3rd | 9. 2nd       | 17. (-1, -1) |
| 2. 2nd | 10. 4th      | 18. (5, -2)  |
| 3. 2nd | 11. 2nd      | 19. (-1, -2) |
| 4. 4th | 12. 3rd      | 20. (3, -1)  |
| 5. 4th | 13. (4, 3)   | 21. (-1, -3) |
| 6. 3rd | 14. (3, -4)  | 22. (4, 4)   |
| 7. 3rd | 15. (-5, -1) | 23. (0, -2)  |
| 8. 2nd | 16. (-1, -5) | 24. (-4, 3)  |

#### Medium (drill)

- |             |                   |                   |
|-------------|-------------------|-------------------|
| 1. (7, -7)  | 11. (8, 7)        | 21. 4th           |
| 2. (3, 3)   | 12. (5, 6)        | 22. 4th           |
| 3. (-4, 2)  | 13. (6, -8)       | 23. 2nd           |
| 4. (-4, 1)  | 14. (-2, -4)      | 24. on the x-axis |
| 5. (-1, 4)  | 15. (-1, 8)       | 25. 3rd           |
| 6. (4, 6)   | 16. (-5, 2)       | 26. 2nd           |
| 7. (-7, -3) | 17. 1st           | 27. 3rd           |
| 8. (-4, 4)  | 18. on the y-axis | 28. the origin    |
| 9. (4, -8)  | 19. 1st           |                   |
| 10. (2, -6) | 20. 2nd           |                   |

#### Challenging

- B: 2nd
- (0, 0)
- (-3, -2)

#### Extension

- 12 square units
- (1, -1), (4, -1), (1, -3)

#### Exit ticket

- Level 1: 2nd
- Level 2: e.g. (3, -1)
- Level 3: (-2, -5)

## 9.04 Plotting tables of values (booklet p. 21)

### WE DO examples (teacher)

1. Ex 1a: vertical line  $x = -2$
2. Ex 1b: horizontal line  $y = 3$
3. Ex 2a: straight line going down
4. Ex 2b: straight line going up steeply
5. Ex 3a:  $y$  goes up by 2
6. Ex 3b:  $y = 7$

### Easy (drill)

- |                                     |             |             |
|-------------------------------------|-------------|-------------|
| 1. (0, 5), (1, 9), (2, 13), (3, 17) | 5. (6, 25)  | 13. (6, 15) |
| 2. (0, 0), (1, 3), (2, 6), (3, 9)   | 6. (4, 16)  | 14. (4, 12) |
| 3. (0, 5), (1, 7), (2, 9), (3, 11)  | 7. (4, 4)   | 15. (3, 14) |
| 4. (0, 2), (1, 3), (2, 4), (3, 5)   | 8. (6, 30)  | 16. (6, 9)  |
|                                     | 9. (3, 16)  | 17. (3, 10) |
|                                     | 10. (3, 5)  | 18. (4, 15) |
|                                     | 11. (3, 15) | 19. (6, 28) |
|                                     | 12. (4, 17) |             |

### Medium (drill)

- |       |        |        |
|-------|--------|--------|
| 1. 24 | 6. 28  | 11. 9  |
| 2. 13 | 7. 3   | 12. 19 |
| 3. 15 | 8. 24  | 13. 8  |
| 4. 19 | 9. 25  | 14. 13 |
| 5. 29 | 10. 10 | 15. 8  |

### Challenging

1. a U-shaped curve (parabola)
2. no, it is a curve

### Extension

1. -5
2.  $y = 4x$

### Exit ticket

1. Level 1: (0, 3), (1, 5), (2, 7)
2. Level 2: horizontal line
3. Level 3: it goes up by 3

## 9.05 Tables of values from rules (booklet p.

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### WE DO examples (teacher)

1. Ex 1a: 2, 3, 4, 5, 6
2. Ex 1b: -11, -9, -7, -5, -3
3. Ex 2a: 6, 5, 4, 3, 2
4. Ex 2b: -5, -2, 1, 4, 7
5. Ex 3a:  $y = 2x + 4$
6. Ex 3b:  $y = 8 - x$

### Easy (drill)

- |       |                   |                   |
|-------|-------------------|-------------------|
| 1. 15 | 8. 5              | 15. 6, 7, 8, 9    |
| 2. 14 | 9. 19             | 16. 6, 10, 14, 18 |
| 3. 16 | 10. 8, 11, 14, 17 | 17. 0, 5, 10, 15  |
| 4. 6  | 11. 4, 8, 12, 16  | 18. 3, 5, 7, 9    |
| 5. 22 | 12. 8, 9, 10, 11  | 19. 9, 10, 11, 12 |
| 6. 2  | 13. 9, 12, 15, 18 |                   |
| 7. 5  | 14. 3, 7, 11, 15  |                   |

### Medium (drill)

- |            |             |              |
|------------|-------------|--------------|
| 1. $x = 2$ | 6. $x = 8$  | 11. $x = 6$  |
| 2. $x = 9$ | 7. $x = 4$  | 12. $x = 7$  |
| 3. $x = 8$ | 8. $x = 1$  | 13. $x = 10$ |
| 4. $x = 1$ | 9. $x = 0$  | 14. $x = 0$  |
| 5. $x = 5$ | 10. $x = 6$ | 15. $x = 9$  |

### Challenging

1.  $y = 2x + 4$
2.  $y = 8 - x$

### Extension

1. 2, 0, 0, 2, 6
2.  $y = x^2$

### Exit ticket

1. Level 1: 3
2. Level 2: -1, 3, 7
3. Level 3:  $y = 3x + 1$

## 9.06 Graphing linear equations (booklet p. 33)

### WE DO examples (teacher)

1. Ex 1a:  $y = -3, -1, 1, 3$ ; straight line
2. Ex 1b: line through (0, 3) going down
3. Ex 2a: yes
4. Ex 2b: yes
5. Ex 3a:  $y = -4, -3, -2, -1, 0$
6. Ex 3b: (0, -1)

### Easy (drill)

- |        |                                       |                                      |
|--------|---------------------------------------|--------------------------------------|
| 1. yes | 9. yes                                | 14. (-1, -2), (0, 1), (1, 4), (2, 7) |
| 2. yes | 10. (-1, -3), (0, -1), (1, 1), (2, 3) | 15. (-1, -1), (0, 0), (1, 1), (2, 2) |
| 3. yes | 11. (-1, -4), (0, -1), (1, 2), (2, 5) | 16. (-1, -3), (0, 0), (1, 3), (2, 6) |
| 4. no  | 12. (-1, 2), (0, 3), (1, 4), (2, 5)   | 17. (-1, -1), (0, 2), (1, 5), (2, 8) |
| 5. yes | 13. (-1, 0), (0, 2), (1, 4), (2, 6)   |                                      |
| 6. yes |                                       |                                      |
| 7. yes |                                       |                                      |
| 8. no  |                                       |                                      |

### Medium (drill)

- |       |         |        |
|-------|---------|--------|
| 1. -4 | 7. 11   | 13. 9  |
| 2. 3  | 8. 2    | 14. 19 |
| 3. 0  | 9. -4   | 15. -1 |
| 4. 4  | 10. -10 | 16. 10 |
| 5. -6 | 11. -1  |        |
| 6. 0  | 12. 12  |        |

### Challenging

1. parallel lines, 2 units apart
2. (1, 2)

### Extension

1.  $m = 3$
2.  $y = 2x + 1$  and  $y = 2x - 3$

### Exit ticket

1. Level 1: 5
2. Level 2: yes
3. Level 3: yes